



Vivaan Shah
Chemical Engineering
Indian Institute of Technology Bombay

23B0330
B.Tech.
Gender: Male
DOB: 24/12/2004

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2027	

Pursuing a Minor in Artificial Intelligence, Machine Learning and Data Science from C-MInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Awarded **AP** grade (**top 1 %**) in **4** courses for exceptional performance & **11 AA's** out of 22 total courses (2025)
- Selected as a **Teaching Assistant** for **MA110** to conduct weekly tutorial sessions for **30+** first year students (2025)
- Secured a perfect **10 SPI** in the third semester of the Chemical Engineering Department across 5 core courses (2024)
- Conferred with the **Professor T.S. Raghunathan prize** for the **Best Student** in UG Thermodynamics (2024)
- Secured **99.78%ile** in JEE Main and **top 1.3%ile** in JEE Advanced out of **12+ lakh** students nationwide (2023)

PROFESSIONAL & ENTREPRENEURIAL EXPERIENCE

AlphaGrep | Quantitative Research Intern (May'25-Jun'25)

- Created a **50+** technical indicator **python library** that uses financial **time-series** data to produce trading signals
- Increased annual returns by nearly **6%** by experimenting with hyperparameter tuning and implementation methods
- Achieved a **2.2 Sharpe Ratio** & **15.3% ARR** by developing modified versions of RSI, ADX and Vortex indicators

Godrej Capital | Data & Risk Analyst Intern (Jun'24-Jul'24)

- Refined the **10,000+** customer base categorization process based on geographical, transactional & behavioural trends
- Improved the accuracy to **91%** of predicting defaulters by utilizing ML methods such as XGBoost, KNN & SVM's
- Implemented the framework for the **optimization** of **15,000+** loan allocations using **Google-OR Tools** & **SciPy**

TicketSure | Entrepreneurship Project (Mar'25-Present)

Event ticket-booking app, aiming to solve India's **₹4,500 Cr+** ticket resale market gap, using a secure resale framework

- Led product development with **4+** key USPs like **dynamic refunds**, incubated at the **Bruce Henderson Ideathon**
- Modeled the **unit economics** with **10-12%** commissions, backed by revenue model analysis of **5+** industry leaders
- Surveyed **120+** users and engaged with **10+** event organizers to validate product-market fit through in-depth research

LEADERSHIP & ORGANIZATIONAL ROLES

Institute Student Mentor | SMP, IIT Bombay (May'25-Present)

Part of a team of **152** mentors aimed at providing academic and personal guidance to incoming first-year students

- One of **only 25** third-year students selected as an ISMP mentor among the entire batch of over **1400+** students
- Selected through a rigorous process comprising of a statement of purpose, interviews and peer and senior reviews

Department Academic Mentor | SMP, IIT Bombay (May'25-Present)

Selected after a rigorous process comprising a statement of purpose, peer reviews and interview performance

- Responsible for mentoring **6 sophomores**, fostering their holistic development and guiding their academic decisions
- Will be responsible for conducting **10+** detailed academic help sessions, events and course reviews for the sophomores

KEY PROJECTS

Bruce Henderson Insight - Ideathon | Boston Consulting Group (Feb'25-May'25)

- Among the top **3** teams representing IIT Bombay in the **national semi-finals (top 30)** out of **1700+** applicants
- Uncovered **15-20%** ticket-waste inefficiency while conceptualizing **TicketSure**, to identify scope for verified resale
- Analyzed India's **₹24k Cr** ticketing market uncovering a **₹800 Cr** growth opportunity using SWOT & market sizing

Digital Modelling & EoS Verification | Course Project | Prof. Guruswamy Kumaraswamy (Sep'24-Nov'24)

- Developed thermodynamic charts & interactive P-V, P-H, T-S diagrams for SF6 using **MATLAB** & **matplotlib**
- Leveraged the **Peng-Robinson** and IV Kudryavtseva EoS' to determine important physical properties such as molar volume, enthalpy, entropy & compressibility by using modified **departure functions** and **reference states**
- Verified the results with **95%** accuracy with Baker & Adamson's findings using MAPE, RMSE & R-value metrics

Marketing Strategy Analysis | Course Project SJM SOM | Guide: Prof. Mayank Pareek (Sep'23-Nov'23)

- Worked in a team of **10+** to analyze boAt's marketing strategy, by studying its **5** competitors and consumer data
- Identified opportunities to enhance marketing strategies by carrying out financial, **PESTEL** and **SWOT** analysis
- Applied the STP and 4P's of marketing, revealing insights such as youth-focused pricing and influencer promotions

EXTRACURRICULAR ACTIVITIES

Football	Captained the IITB football team in Aavhan, securing the 3rd position out of 18 total teams - Selected for the Inter-IIT Football Camp and represented IITB at Skream and Spree 2025 - Played in global tournaments like FC Bayern Youth Cup & Costa Blanca Cup in Spain - Represented the school football team for 10+ years winning tournaments like MSSA & Jesuits
Literary & Debate	- Conferred with the Outstanding Delegate award of DISEC at Harvard MUN India 2020 - Recognised for the Verbal Mention award of HSC at Cathedral Model United Nations 2020 - One of 35 Star Correspondents out of 1k participants in the TOI NIE Journalism Program
Miscellaneous	- National Finalist (Top 30) out of 20,000+ in the Innoventure Entrepreneurship Competition - Taught Math to the underprivileged through NGO's like Social Connect & Angel Xpress